

List in Python – Team Manager:

MAKE A LIST

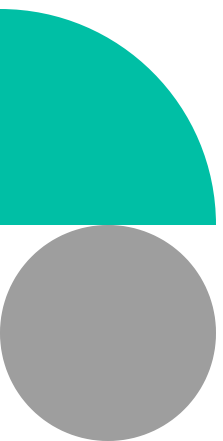
Informatics – Grade 8



Learning Objective



- 01 Understand List Variable
- 02 Understand List Variable
- 03 Identify elements in a list.



What is Variable?

You learned that a variable stores a value.

For example:

```
number = 5  
colour = "red"  
name = "Ade"
```

variable like a container or box that stores data.

We can think of it like a box:



What is List?

- List is a special type of variable
- List can store several different data items
- List is an ordered and mutable collection data type.
- This data type can be defined using square brackets `[]` in Python.
- Elements are separated by commas.

mouse
(data)



box(List)

List in Python

for example:

```
colourlist = ["red", "yellow", "blue", "green"]  
print(colourlist)
```

output:

```
['red', 'yellow', 'blue', 'green']
```

This called is element



Append an Element

Append means 'add to the end'. You can append items to a list. That means a new item will be added to the list. This command will append "orange" to colourlist.

code: `colourlist.append("orange")`

full code :

```
colourlist = ["red", "yellow", "blue", "green"]  
print(colourlist)  
colourlist.append("orange")  
print(colourlist)
```

output :

```
['red', 'yellow', 'blue', 'green']  
['red', 'yellow', 'blue', 'green', 'orange']
```

Append user input

You can get input from the user and append that value to the list.

```
colour = input("Enter a colour ")  
colourlist.append(colour)
```

full code :

```
colourlist = ["red", "yellow", "blue", "green"]  
print(colourlist)  
colour = input("Enter a colour ")  
colourlist.append(colour)  
print(colourlist)
```

output:

```
['red', 'yellow', 'blue', 'green']  
Enter a colour orange  
['red', 'yellow', 'blue', 'green', 'orange']
```

Append elements using a loop

A loop repeats commands.

Python has two common types of loops:

- ◆ For loop

Use a for loop when we know how many times we want to repeat.

- ◆ While loop

Use a while loop when we want to repeat while a condition is true.

Using a for Loop

Imagine our team has 5 players.

We know exactly how many names we need. So we can use a for loop. example code:

```
teamlist = []  
for i in range(5):  
    name = input("Enter a player name: ")  
    teamlist.append(name)  
print(teamlist)
```

output:

```
Enter a player name: Ade  
Enter a player name: Ole  
Enter a player name: Mitch  
Enter a player name: Nadeo  
Enter a player name: Sandy  
['Ade', 'Ole', 'Mitch', 'Nadeo', 'Sandy']
```

Activity

Task 1 – Run the Program

1. Create an empty list called teamlist.
2. Use a for loop to ask the user to enter player names.
3. Add each player name to teamlist.
4. Run the program and enter the player names.
5. Display the list after all names have been entered.
6. Check that all player names have been added to the list.

Using a while Loop

The loop repeats while the test is True, and stops when it is False. It is controlled by the question "Do you want to add another? (Y/N)". If the user types y, the loop repeats. Anything else stops it. example

code:

```
teamlist = []
repeat = "Y"
while repeat == "Y":
    name = input("Enter a player name: ")
    teamlist.append(name)
    repeat = input("Add another player? (Y/N): ")
print(teamlist)
```

output:

```
Enter a player name: Ade
Add another player? (Y/N): Y
Enter a player name: Jay Idzes
Add another player? (Y/N): N
['Ade', 'Jay Idzes']
```

Activity

Task 2 – Run the Program

1. Create an empty list called teamlist.
2. Use a while loop to ask the user to enter a colour.
3. Add each colour to teamlist.
4. After entering a colour, ask the user if they want to add another colour.
5. Continue the program if the user enters Y.
6. Stop the program if the user enters N.
7. Display the list after the program stops.
8. Check that all the colours have been added to the list.



Thank You
For Your Attention

